

EFFECTS OF ROOT GROWTH AND DECAY ON THE PERMEABILITY OF A SYNTHETIC SANDY LOAM

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From field experiments on perennial pastures grown on a sandy loam, the author² found that, as the concentration of macroorganic matter in the top 3 inches of soil increased, the infiltration rate decreased. It was suggested that blockage of pores by underground plant parts might account for the decrease in infiltration rate, and the following laboratory experiment was designed to test this hypothesis.

EXPERIMENTAL PLAN

Four columns of 250 cc. volume were packed with organic-matter-free, non-aggregated sandy loam synthesized from quartz sand and kaolin clay. Intrinsic permeability of each column at saturation (P_1) was measured under sterile conditions. Two of the columns were each sown with 14 corn seeds, and two columns were left as controls. Ten days later, measurement of permeability (P_2) was begun.

The corn was grown and P_2 was measured under sterile conditions in one run of the experiment, and under nonsterile conditions in a separate run in which the columns were inoculated at sowing time with 1 ml. of a suspension of 3 g. of garden soil in 1 l. of water.

After determination of P_2 , plant tops were harvested, oven-dried, and weighed. Fresh root volume was found for one planted column by direct measurement, and was estimated for the second from shoot weight, the ratio of shoot weight to root volume of the first column being used. After harvesting of tops, this second planted column and a control were inoculated and incubated at room temperature. Permeability (P_3) was measured after 1 month's incubation, and P_3' was measured after 2 months' incubation.

Pore space was measured after determination of P_1 and of P_2 . The paths of movement of permeating Ponceau red solutions were observed. The effects of root growth on pore-size distribution were studied microscopically on polished sections of "Castolite"-impregnated columns.

APPARATUS AND METHODS

The permeameters were constructed from Pyrex filter mantles of 6 cm. internal diameter, sectioned at 10 cm. from the mantle base. Piezometer tubes were

¹ Experiment conducted at the Agronomy department, Cornell University, Ithaca, N. Y., by the author while on leave from C.S.I.R.O., Deniliquin, Australia. Grateful acknowledgment is made to Dr. R. D. Miller, who advised on all phases of the experiment, and to Dr. F. Broadbent for advice on microbiological techniques.

² Barley, K. P. The root growth of irrigated perennial pastures and its effects on soil structure. *Australian J. Agr. Research* 4: 283-291. 1953.

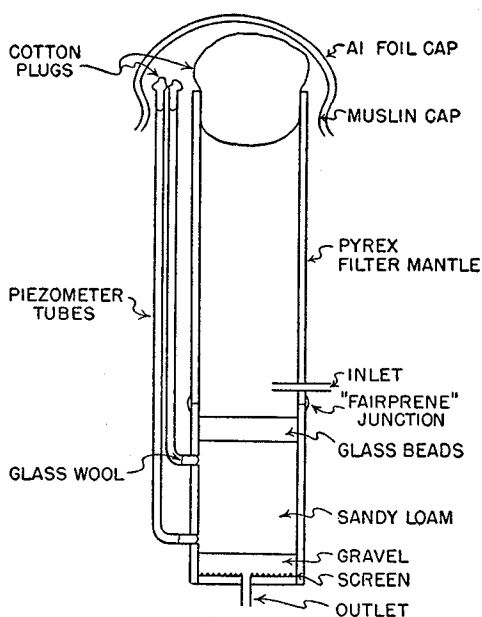


FIG. 1. PERMEAMETER DESIGN

attached 2.5 cm. and 7.5 cm. above the mantle base. The permeameter design is shown in figure 1. *Nutrient solution* was drawn through a battery of porcelain Pasteur-Chamberlain microbial filters into a 20-l. reservoir. Hoagland and Arnon³ No. 2 nutrient solution was used as the permeating fluid throughout the experiment. The filtered solution was siphoned into 4-l. reservoirs, which were equipped with constant-head Mariotte tubes and which supplied the permeameters. Cotton gauze filters dusted with fungicide were placed in the aspirator and Mariotte-tube air lines and at all other air entry points. Outflow from the permeameters was led through mercuric-chloride trap bottles into 100-ml. measuring cylinders. The layout of the apparatus is shown in figure 2.

Sandy loam was prepared by dry-mixing four parts of quartz sand with one part of kaolin clay, each fraction being free of organic matter. Each permeameter was packed by adding 35 g. of coarse sand, to form a layer adjacent to the soil-retaining screen, followed by 370 g. of sandy loam, the soil column being compacted by dropping the unit 20 times through a height of 3 cm. The top of each column was leveled and covered with 35 g. of 3 mm. diameter glass beads. The soil columns were sterilized by autoclaving at 15 pounds for 5 hours. Glassware, porcelain-filters, and rubber connections were autoclaved for 1.5 hours. All connections were wrapped with cotton-gauze and muslin before autoclaving. Corn seeds were sterilized by washing under reduced pressure in ethyl alcohol for 1 minute and in 0.2 per cent HgCl_2 for 5 minutes, followed by four rinsings with sterile water. Soil sterility was checked by inoculating "Difco-Bacto"

³ Hoagland, D. R., and Arnon, D. I. The water-culture method or growing plants without soil. *Calif. Agr. Expt. Sta. Circ.* 347: 1-39. 1938.

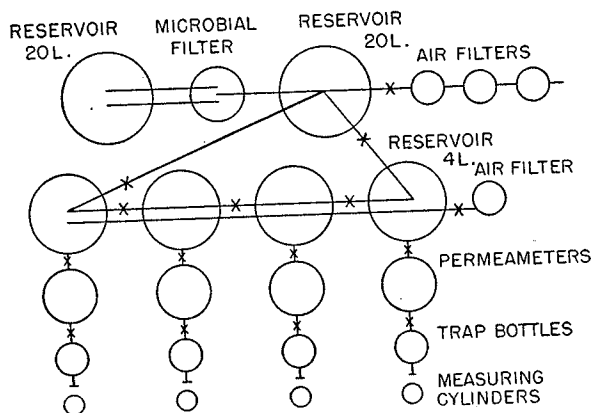


FIG. 2. LAYOUT OF APPARATUS

nutrient broth with loop samples and incubating for 1 week at room temperature. Plate counts of contaminants were made on albumin agar and dextrose peptone agar according to the methods of Waksman.⁴

The determination of *permeability* (P) required construction of permeability-time curves. In sterile runs, permeability reached a stable value within 100 to 150 hours, this value being taken as P . In nonsterile runs, permeability passed through a maximum, which was taken as (P) . Hydraulic gradients were maintained in the range 0.5 to 1.0. *Pore space* was calculated from the mass and measured dimensions of the sandy loam columns. *Root volume* was measured directly by displacement of water, the roots having been separated from soil by washing on a 1-mm. screen, cleaned by shaking in water, and freed from excess water by brisk shaking in air. *Polished sections* of sandy loam through which corn roots had been grown were prepared according to the method of Day,⁵ except that sections were cut with a diamond saw. With an ocular scale at $\times 60$ magnification, pore intercepts were measured on 0.9-mm. transects adjacent to root channels and on transects more than 5 mm. from the nearest root.

RESULTS

Effects of root growth on permeability and pore space

Sterile run. Microorganisms were not excluded completely, 3000 bacteria per gram of soil being present at the end of the second permeability determination. Fungi were absent. It is unlikely that this population had any measurable effect on permeability or pore space. Table 1 records changes produced during the growth of corn roots.

Nonsterile run. Table 2 records changes produced during the growth of corn roots under nonsterile conditions.

⁴ Waksman, S. A. *Principles of Soil Microbiology*, pp. 16-20. Williams & Wilkins Co., Baltimore. 1927.

⁵ Day, P. R. Experiments in the use of the microscope for the study of soil structure. *Soil Sci. Soc. Amer. Proc.* (1948) 13: 43-49. 1949.

TABLE 1
Permeability and pore space changes produced by growth of corn roots under near-sterile conditions

	Permeability	Total Pore Space	Root-Free Pore Space
Mean value before sowing (stage 1).....	1.66 μ^2	43.3%	43.3%
	%	%	%
Change for each column, per cent of the stage 1 value			
Corn.....	-89	+0.9	-9.3
Corn.....	-91	+1.2	-8.1
Not sown.....	+8	+0.9	+0.9
Not sown.....	+16	+0.9	+0.9

TABLE 2
Permeability and pore space changes produced by growth of corn roots under nonsterile conditions

	Permeability	Total Pore Space	Root-Free Pore Space
Mean value before sowing (stage 1).....	2.36 μ^2	41.2%	41.2%
	%	%	%
Change for each column, per cent of the stage 1 value.....			
Corn.....	-92	-0.4	-8.4
Corn.....	-95	+0.8	-8.4
Not sown.....	+14	-0.4	-0.4
Not sown.....	-23	0.0	0.0

The large and consistent decrease in permeability of the columns sown to corn may be attributed to reduction of root-free pore space, the volume increase of the roots having failed to cause a corresponding increase in apparent volume of the soil column. Since the permeability drop occurred when the roots were growing in the presence of a very sparse microbial population, the reduction was essentially due to the physical effects of the roots themselves. The permeability of controls varied between stage 1 and stage 2, the most likely causes of variation being gradual closure of cracks that had developed during autoclaving and differences in the completeness of saturation. The placement of piezometers 1.5 cm. above the base and 1.5 cm. below the top of the sandy loam column eliminated variation due to seed placement, crowding of roots at the column base, and surface dispersion.

Dissection of columns infiltrated with Ponceau red solution showed no regular difference between flow velocity at root/soil boundaries and at points within the soil. The roots were, therefore, in close contact with their channel boundaries. Flow velocity of the dye solution was greater at the glass wall interface than within the column and was greater at root/glass boundaries than at soil/glass boundaries. These aberrant flow conditions at the glass boundary opposed, but did not mask, main treatment effects.

TABLE 3
*Effects of root growth on pore space and pore size distribution**

	$\frac{\Delta \text{ Pore Length}}{\text{Transect Length}} \cdot 100$	Per Cent of Total Pore Length in Interval			
		< 0.05 mm.	0.05-0.09 mm.	0.09-0.14 mm.	> 0.14 mm.
0.9 mm. transect adjacent to root..	33.1	13.2	32.8	26.8	27.2
0.9 mm. transect > 5 mm. from nearest root.....	38.7†	12.1	28.1	28.1	31.7

* Minute pores within granular bodies and clay coatings were not measurable by this method.

† S.E. = 2.68.

TABLE 4
Permeability changes during root growth and decay

	Corn	Not Sown
	μ^2	μ^2
Before sowing (P_1).....	1.87	1.81
After 15 days' growth (P_2).....	0.21	1.97
After 1 month's decay (P_3).....	0.99	2.06
After 2 months' decay (P_3').....	1.25	1.91

Measurements of pore intercepts on optical transects revealed that soil adjacent to root channels of 1 mm. diameter was more compact than soil several millimeters removed. Mean values for 30 pairs of transects are recorded in table 3. Sand grains in transects adjacent to root channels were disposed with flat faces rather than corners in contact, and with flat faces parallel to channel boundaries. Roots of 1 mm. diameter formed straight channels. Finer roots, of 0.1 mm. diameter, also displaced soil grains but underwent some deflection, forming gently curved channels.

Effects of root decay on permeability

After 1 month's decay (stage 3) at room temperature, cortical tissue had disintegrated. Epidermal and endodermal rings of tissue remnants and many vascular elements remained. Mycelium was prominent within the endodermis growing on vascular remnants; but microbial growth did not block the space left by cortical disintegration. After 2 months' decay (stage 3'), tissue remnants and the associated microflora remained as poorly coherent coatings on the channel walls.

Permeability data for a treated unit and a control at all stages of the experiment are recorded in table 4. Permeability decreased to a low value during the growth of the corn but increased during the period of decay. The extent of the increase during decay may be expected to vary widely between units, depending on the channel continuity provided between piezometer limits.

* See footnote 2.

DISCUSSION

The growth of roots through sandy loam has been shown to reduce permeability. For the single grained synthetic soil used in the laboratory experiment, the permeability drop was due chiefly to root pressure having displaced and rotated soil particles into a more compact and more mechanically stable arrangement. For structured soils, blockage of existing cleavages and channels by growing roots may be of greater importance.

Field experiments on 3-year-old pastures grown on a sandy loam⁶ showed that infiltration rate and concentration of macroorganic matter in the top 3 inches of soil were inversely related between concentration limits of 0.6 and 2.2 g. per 100 g. The relationship cannot be extrapolated to lower concentrations, since the decrease in herbage growth associated with decrease in macroorganic matter concentration leads to surface exposure and a fall in permeability. In the laboratory experiment the roots grew through continually moist soil of negligible organic matter content. These special conditions favored particle rearrangement by root pressure, but did not favor formation of stable aggregates, which may be chiefly shrinkage structures bonded by condensed decomposition products. In the field experiment, however, the plants were grown for 3 years through repeated wetting and drying cycles.

The data obtained qualify numerous earlier observations of structure build-up under sod, which influences structure by, for example, prevention of surface puddling and enhancement of earthworm activity, as well as by root growth and decay. In the field, under sod, the following tendencies will operate: the tendency of root growth to decrease permeability through pore blockage and compaction; the opposing tendency of root decay to increase permeability through the freeing and stabilization of channels for water movement. The level of permeability may shift with seasonal or induced change in the relative strengths of the opposed processes of growth and decay.

SUMMARY

The growth of corn roots through a column of synthetic sandy loam, under either sterile or nonsterile conditions, compacted the soil between the root channels and greatly reduced the permeability of the column. When the roots decayed, the freeing of their channels for water movement caused an increase in permeability.